

VAISHALI MD

B.Tech Information Technology | Sri Muthukmaran Institute of Technology, Tamil Nadu

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WORK EXPERIENCE - 9.6YRS

Associate Data Scientist (*unpaid*) | ProITBridge PVT LTD, Bangalore **March 2025 - Present**

- Designed and implemented a comprehensive Recommendation System for movies, TV series, documentaries, and music, resulting in a Boosted 17% in customer engagement.
- Designed and implemented a cutting-edge Quality Check System integrating YOLO (You Only Look Once) model for precise image processing. This solution significantly accelerated the inspection process, achieving an efficiency boost of 0.6 times, (60%) enhancing the detection and quantification of various holes on metal plates for improved quality control.
- Building and fine-tuning machine learning models for various tasks such as classification, regression, clustering, and recommendation.
- Identifying and engineering relevant features from raw data to improve model performance.

Senior QA | FEI Systems PEO Model,US

June 2022 - July 2024

- Designed and implemented test strategies, test plans, and test cases for functional, regression, and performance testing.
- Spearheaded end-to-end testing for complex software systems, ensuring high-quality releases with 50% pass rate.
- Manage and prioritize multiple projects simultaneously, meeting deadlines and delivering high quality results.
- Collaborating with cross-functional teams, including business analysts, IT professionals, and management, to understand data needs and requirements.

Senior System Engineer | IBM India PVT LTD, Chennai

February 2015 - June 2022

- Conducted manual testing (functional, UI, cross-browser) for web and mobile applications.
- Developed and maintained automated test scripts in Java/Python/JavaScript, increasing test coverage by 30%.
- Performed API testing (Postman, REST Assured) and validated backend services for reliability and scalability. Executed load and performance testing using JMeter/LoadRunner, identifying bottlenecks in high-traffic scenarios.
- Worked in Agile/Scrum environments, participating in sprint planning, backlog grooming, and defect triage. Improved bug tracking and reporting using JIRA/Xray, ensuring clear communication between QA and development teams.

PROJECTS

EHR Data Quality Improvement Initiative

- Conducted comprehensive analysis of 500K+ patient records to identify 12 recurring data quality issues (e.g., duplicate records, missing insurance info). This facilitates further investigation.
- Developed automated data validation scripts that reduced data entry errors by 38%.
- Implemented regression test suite specifically targeting high-risk data fields identified through analysis.
- Involved in the Exploratory Data analysis, Prediction with machine learning, Report analysis with chart in python. Data analysis goes beyond prediction. It delves into sensor data, usage patterns, and environmental factors to pinpoint the root causes of component failures.

Fraud Detection System Testing

- Conducted exploratory testing to identify gaps in fraud rule logic, uncovering 12+ edge cases missed in requirements.
- Performed UAT with business analysts to validate fraud alerts against real-world transaction patterns.
- Used SQL queries to verify data consistency between fraud alerts and backend transaction logs.
- Built Python scripts to analyze 2M+ transactions and auto-generate test data for fraud scenarios. Automated API tests (Postman) for real-time fraud scoring, reducing regression time by 50%.

Healthcare Claims Processing Platform

- Analyzed business requirements and user stories to identify testing needs and define acceptance criteria.
- Created over 200+ detailed test cases for modules like claim submission, adjudication, rejection, and EOB generation.
- Performed end-to-end functional testing of claims lifecycle including ICD/CPT code validation and payer-provider rules.
- Executed regression tests during every sprint and major release using a risk-based testing approach.
- Conducted API testing using Postman to validate response payloads, status codes, and claim data flow between services.
- Participated in daily stand-ups, grooming sessions, and sprint planning to align QA timelines with development.
- Collaborated with developers and product managers to troubleshoot critical issues, resulting in 35

Telecom Customer Self-Service Portal

- Analyzed business requirements and created detailed test cases for modules including user registration, prepaid/postpaid recharge, data usage tracking, complaint management, and service activation.
- Validated user journeys for SIM activation, mobile number portability (MNP), and plan upgrades. Conducted backend validation using SQL to verify recharge history, data usage logs, and complaint tickets.
- Reported and tracked over 120+ issues using Jira and ensured timely retesting and closure before sprint deadlines.
- Participated in Agile ceremonies including sprint planning, daily stand-ups, and retrospectives.

EDUCATION

B.Tech Information Technology Sri Muthukumaran Institute of Technology, Chennai, Tamil Nadu (2010–2014)

SKILLS

• Python	• TensorFlow	• Manual Testing	Execution
• SQL	• PyTorch	• Functional Testing	• Cross-browser
• Pytest	• Scikit-learn	• Regression Testing	Cross-device Testing
• Unittest	• MLflow	• UI/UX Testing	• SQL for Data
• Selenium	• Jenkins	• Test Planning	Validation
• Postman	• GitHub Actions	Strategy	• XGBoost , LightGBM
• ML Model Validation	CI/CD	• Test Case Design	• API Testing

TECHNICAL SKILLS

Languages: Python, SQL, Docker, Javascript, Java, Gherkin, XML, HTML/CSS
Technologies/Frameworks: Pytest, Unittest, Selenium, Behave, Scikit-learn, TensorFlow, PyTorch, MLflow, DVC, Apache Airflow, Jenkins Pipelines, GitHub Actions
Tools: Tools: Pytest, Unittest, Selenium, Postman, MLflow, Evidently AI, DVC, TensorBoard, Jenkins, GitHub Actions, CircleCI, Docker, Git, TestRail, Prometheus, Grafana

ACHIEVEMENTS

- Improved test coverage by 60 percentage by designing detailed test cases for complex telecom workflows.
- Received Employee of the Year award for outstanding performance.
- Led UAT for a major healthcare release with 100 percentage stakeholder satisfaction and zero post-go-live issues.

DECLARATION

- This resume truthfully reflects my qualifications and experience.